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DENSITY-DEPENDENT NATURAL SELECTION¹

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Abstract. Density-dependent selective values illustrate the evolutionary effect of population-regulating processes that diminish an individual's probability of survival with increased crowding. The selective values, assumed to decrease as a linear function of density, lead in a mild environment to the evolution of phenotypes having a high carrying capacity, K , at the expense of a low intrinsic rate of increase, r . A graphical technique shows that selection causes evolution of phenotypes having a high r at the expense of a low K in harsh seasonal environments.

A mathematical technique developed for analyzing evolution in coarse-grained seasonal environments reveals genetic mechanisms, including ones with full dominance, with which a moderately harsh seasonal environment causes stable polymorphism between high- r and high- K genes.

The energy balance equation demonstrates the role of high- r and high- K phenotypes in the population's energy flow. A high- r phenotype makes a large expected contribution to the population's productivity under conditions of negligible crowding, and a high- K phenotype has, for a given contribution to the population's productivity under uncrowded conditions, a low sensitivity to having that contribution diminished by crowding.

Ecologists (e.g., Birch 1960) have recognized that natural processes that cause population regulation also effect, by way of natural selection, the qualitative properties of the organisms in the regulated populations. Many of these regulatory processes diminish an individual's probability of survival with increasing population density, while others diminish an individual's expected fertility. This paper examines the evolutionary effect of population-regulating processes that diminish the individual's probability of survival. Clearly, for any natural selection to occur at all, the processes must diminish the probabilities of survival to a different extent for various genotypes present in the population.

Early theoretical work making reference to density effects in evolution was done by MacArthur (1962) and by MacArthur and Wilson (1967). These authors form a distinction between two kinds of population genetics, one in which fitness is defined as r , the intrinsic rate of increase, and one in which fitness is defined as K , the carrying capacity. They assert that the population genetics in which fitness is the intrinsic rate of increase applies to expanding populations under conditions of negligible crowding where population density has no influence on birth and death rates; " r -selection" is said to be occurring in such a situation. The population genetics in which fitness is the carrying capacity applies to crowded populations where " K -selection" is said to be occurring. Also, MacArthur and Wilson write "where climates are rigorously seasonal and winter survivors recolonize each spring, we expect r -selection . . . ; where climates are uniformly benign, K -selection" In addition, they interpret the result of r -selection to be the evolution of phenotypes having high pro-

ductivity, and of K -selection, phenotypes having high efficiency.

This present paper differs from and extends the earlier work of MacArthur in certain ways. First, the fitness is throughout measured by the selective value introduced by Sewall Wright. The analysis shows r -selection to occur in situations where phenotypes possessing high r while sacrificing K have the highest selective value, and K -selection to occur in situations where phenotypes possessing high K while sacrificing r have the highest selective value. Evolution in every case favors the genes producing phenotypes with the highest selective value, and hence one need not think there exist two (or more) kinds of population genetics each with its own definition of fitness. Second, this paper is primarily concerned with evolution in environments of intermediate harshness. It establishes the environmental conditions leading to stable coexistence between high- r and high- K genes, and it determines what happens in the region of environmental harshness marking the transition between r -selection and K -selection. Third, the study shows that K -selection results in phenotypes characterized not by a high efficiency, but by a low sensitivity to having their productivity diminished by crowding; however, the result of r -selection is indeed phenotypes with high productivity.

SIMULTANEOUS EQUATIONS FOR ΔN AND Δp

In this section we derive simultaneous equations to predict both the change in population size and gene pool composition at one locus from measurements of a single set of parameters. The equations are essentially those of Sewall Wright, and the reader familiar with his work may skip to the next section. We assume the population under study has a life

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history clearly divided into discrete generations. During any generation the population consists initially of zygotes, then of immature individuals, and finally of adults. Near the end of their lives the adults interbreed and the resultant crop of zygotes initiates the next generation. We chose to census the population at the time of interbreeding, as it seems more interesting to talk of the properties of adults, but another possible choice is to census the zygotes. The population is potentially polymorphic for a trait determined at one autosomal locus with two alleles, A and B. When the interbreeding occurs the mating is random with respect to the locus of interest.

The set of basic symbols is defined below:

- N^i = total number of adults at time of interbreeding for i th generation
 N_{AA}^i = number of adults with AA genotype at time of interbreeding for i th generation
 N_{AB}^i = (similar to above)
 N_{BB}^i = (similar to above)
 $2m$ = average number of zygotes resulting from a mating—assumed to be independent of the parents' genotype at the locus of interest
 l_{AA} = average fraction of zygotes with AA genotype surviving to take part in breeding
 l_{AB} = (similar to above)
 l_{BB} = (similar to above)

Note especially that the superscript i denotes the generation. It is *not* an exponent.

Since m , the average individual fertility, is independent of the genotype, selection occurs through differential survival only. Both m and the genotypic specific survival fractions, l_{AA} , etc., will in general depend on population size and gene pool composition; they are not constants. Since there are only three genotypes at the locus of interest,

$$N^i = N_{AA}^i + N_{AB}^i + N_{BB}^i \quad (1)$$

The gene pool composition is described by the frequency of the A allele, p , or equivalently of the B allele, q . By definition,

$$p^i = \frac{2N_{AA}^i + N_{AB}^i}{2(N_{AA}^i + N_{AB}^i + N_{BB}^i)} \\ q^i = \frac{2N_{BB}^i + N_{AB}^i}{2(N_{AA}^i + N_{AB}^i + N_{BB}^i)} \quad (2)$$

The frequencies of the various genotypes are defined as:

$$D^i = \frac{N_{AA}^i}{N^i}, \quad H^i = \frac{N_{AB}^i}{N^i}, \quad R^i = \frac{N_{BB}^i}{N^i} \quad (3)$$

Clearly,

$$p^i + q^i = 1, \quad D^i + H^i + R^i = 1 \\ p^i = D^i + (\frac{1}{2}) H^i, \quad q^i = R^i + (\frac{1}{2}) H^i \quad (4)$$

Table 1 presents the steps in deriving the numbers of each genotype present in the $(i+1)$ th generation from the population size and genotypic frequencies in the i th generation. The first column tabulates the number of matings between all possible pairs of parental genotypes, assuming random mating. The first entry in this column is formed, for example, as follows: the probability that any arbitrary mating in the i th generation is between an AA parent and another AA parent, where the parents are chosen at random, is $(D^i)^2$. Since there are N^i organisms, and each mates on the average once, there are $N^i/2$ total matings. Then the expected number of matings between two AA parents is $\frac{1}{2}(D^i)^2 N^i$. A similar argument is made for the other entries in the column. The progression through the remaining columns is self-explanatory.

One will observe from the last column in Table 1 that the parental fertility and the various offspring survival fractions appear together as a product. This fact allows definition for each genotype, say AA, of a number designated by $W_{AA}(N^i, p^i)$ and called the selective value of the phenotype expressed by the AA genotype in the i th generation. The selective values are defined according to

$$W_{AA}(N^i, p^i) = l_{AA}(N^i, p^i)m(N^i, p^i) \\ W_{AB}(N^i, p^i) = l_{AB}(N^i, p^i)m(N^i, p^i) \\ W_{BB}(N^i, p^i) = l_{BB}(N^i, p^i)m(N^i, p^i) \quad (5)$$

The selective value is a very useful measure of the fitness of the phenotype expressed by a given genotype. (The selective value in this paper is the "absolute selective value" of Wright 1959, p. 112). The selective value is empirically determined, as the definition indicates, from the product of the average parental fertility and an average offspring survival fraction. Statistical analysis of data for any particular situation will usually show both the parental fertility and offspring survival fractions, and therefore the selective value, to be functions of the parental population size and gene pool composition. The entire range of selective values for various parental population sizes and gene pool compositions is given by a function of two variables: $W_{AA}(N, p)$, which, for application to any generation, say j , is evaluated at $N = N^j$ and $p = p^j$. Birch (1954) presents evidence for genotypes in *Drosophila* where the selective values are functions of density.

Using the last column of Table 1 and the definition of the gene frequency, p , from (2), we obtain

$$p^{i+1} = \frac{p^i W_{AA}(N^i, p^i) + q^i W_{AB}(N^i, p^i)}{\bar{W}(N^i, p^i)} p^i \quad (6)$$

where

$$\bar{W}(N^i, p^i) = (p^i)^2 W_{AA}(N^i, p^i) + 2p^i q^i W_{AB}(N^i, p^i) \\ + (q^i)^2 W_{BB}(N^i, p^i) \quad (7)$$

TABLE 1. Derivation of the numbers of breeding adults of each genotype in the $(i + 1)$ th generation from the genotypic proportions and population size in the i th generation

Parental genotypes	Average number of matings between given parental genotypes in i th generation	Average number of zygotes from matings of given parent combinations	Number of zygotes of various genotypes according to Mendelian ratios		
			AA	AB	BB
AA $\times$ AA	$(\frac{1}{2}) (D^i)^2 N^i$	$(D^i)^2 N^i m$	$(D^i)^2 N^i m$		
AA $\times$ AB	$(\frac{1}{2}) 2D^i H^i N^i$	$2D^i H^i N^i m$	$D^i H^i N^i m$	$D^i H^i N^i m$	
AB $\times$ AB	$(\frac{1}{2}) (H^i)^2 N^i$	$(H^i)^2 N^i m$	$(\frac{1}{4}) (H^i)^2 N^i m$	$(\frac{1}{2}) (H^i)^2 N^i m$	$(\frac{1}{4}) (H^i)^2 N^i m$
AA $\times$ BB	$(\frac{1}{2}) 2D^i R^i N^i$	$2D^i R^i N^i m$		$2D^i R^i N^i m$	
AB $\times$ BB	$(\frac{1}{2}) 2H^i R^i N^i$	$2H^i R^i N^i m$		$H^i R^i N^i m$	$H^i R^i N^i m$
BB $\times$ BB	$(\frac{1}{2}) (R^i)^2 N^i$	$(R^i)^2 N^i m$			$(R^i)^2 N^i m$
Offspring genotypes	Total number of zygotes		Number of breeding adults in $(i + 1)$ th generation		
AA	$N^i m [(D^i)^2 + D^i H^i + (\frac{1}{4}) (H^i)^2] = N^i m [D^i + (\frac{1}{2}) H^i]^2 = N^i m (p^i)^2$		$l_{AA} m (p^i)^2 N^i$		
AB	$N^i m [D^i H^i + (\frac{1}{2}) (H^i)^2 + 2D^i R^i + H^i R^i] = N^i m 2 [D^i + (\frac{1}{2}) H^i] [R^i + (\frac{1}{2}) H^i] = N^i m 2 p^i q^i$		$l_{AB} m 2 p^i q^i N^i$		
BB	$N^i m [(R^i)^2 + H^i R^i + (\frac{1}{4}) (H^i)^2] = N^i m [R^i + (\frac{1}{2}) H^i]^2 = N^i m (q^i)^2$		$l_{BB} m (q^i)^2 N^i$		

Similarly, using the last column of Table 1 and equation (1) we obtain:

$$N^{i+1} = \bar{W}(N^i, p^i) N^i \quad (8)$$

Equations (6) and (8) can be rewritten as

$$\Delta p^i = p^{i+1} - p^i = \frac{p^i W_{AA}(N^i, p^i) + q^i W_{AB}(N^i, p^i) - \bar{W}(N^i, p^i)}{\bar{W}(N^i, p^i)} p^i \quad (9)$$

$$\Delta N^i = N^{i+1} - N^i = [\bar{W}(N^i, p^i) - 1] N^i \quad (10)$$

Equations (9) and (10) are the simultaneous equations for the change in population size and gene frequency over a generation. An extremely important property of these equations is that measurements of a single set of parameters, the selective value functions, $W_{AA}(N, p)$, $W_{AB}(N, p)$ and $W_{BB}(N, p)$, enable prediction of both ΔN and Δp .

One of the interesting uses for the equations predicting gene frequency and population size is in determining trajectories on the phase plane. The phase plane is a graph whose horizontal axis is scaled for p and vertical axis for N . A trajectory indicates the entire change in N and p over many consecutive generations starting from some initial values, N^0 and p^0 . Clearly, an entire trajectory is produced by successive applications of equations (6) and (8) starting with N^0 and p^0 .

THE MODEL

In the last section the selective values, e.g., $W_{AA}(N, p)$, were regarded as being density and frequency dependent without any additional assumption specifying the functional form of that dependence. To assign some particular function as a selective

value is to invoke a particular model of density- and frequency-dependent selective pressures. In this section the selective values will express what seems the simplest model of density-dependent selection. The selective values are assumed to drop off linearly with population size, as in the equation $W(N) = a - bN$. But instead of writing the parameters as a and b , we write $(r + 1)$ and (r/K) , for reasons that will be clear shortly. Thus the selective values are:

$$\begin{aligned} W_{AA}(N) &= (r_A + 1) - (r_A/K_A)N \\ W_{AB}(N) &= (r_H + 1) - (r_H/K_H)N \\ W_{BB}(N) &= (r_B + 1) - (r_B/K_B)N \end{aligned} \quad (11)$$

$r_A > r_B, K_B > K_A$

Refer in Fig. 1 to the graphs of $W_{AA}(N)$ and $W_{BB}(N)$. At low population sizes the AA homozy-

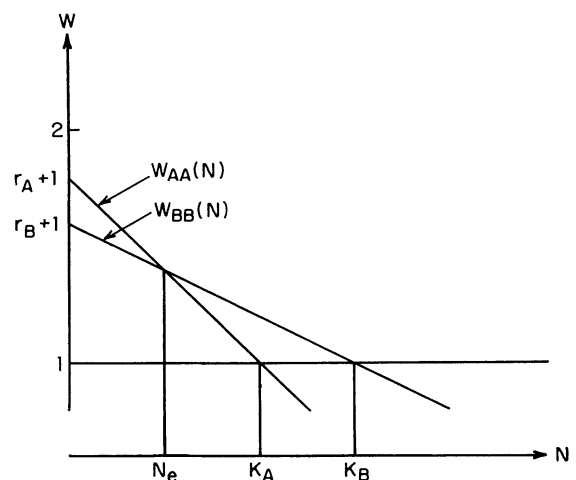


FIG. 1. Selective value of the homozygote phenotypes as a function of population density.

gotes are more fit while at high population sizes the BB homozygotes are more fit. The intercepts are labeled in terms of r and K . A is the high- r allele, and B the high- K allele. N_e designates the population size at which the selective values of both homozygotes are equal.

The selective values determine simultaneously both gene pool evolution, Δp , and population growth, ΔN . The meaning of the parameters, r and K , in the homozygote selective value functions, $W_{AA}(N)$ and $W_{BB}(N)$, can be deduced by considering the population growth entailed by these functions if the locus is fixed for either allele. Suppose the locus is fixed for the A allele. Then substituting $p = 1$, and the selective value functions, (11), into the equation for ΔN , (10), yields

$$\Delta N = r_A N \frac{(K_A - N)}{K_A} \quad (12)$$

Equation (12) describes logistic growth with r_A as the intrinsic rate of increase, and K_A the carrying capacity. A similar argument exists for r_B and K_B . Fig. 2 illustrates growth curves for populations consisting entirely of AA homozygotes or BB homozygotes. As illustrated, the growth curves predicted by the logistic difference equation lead to an asymptotically stable equilibrium at $N = K$ provided that r is between 0 and 1, and hence we will restrict r to within (0,1). (See Smith 1968, p. 26.) A selective value thus has two complementary interpretations. The first, of special relevance to evolutionary biology, is as a measure of the fitness of a phenotype; the second, of special relevance to population dynamics, is as a determinant of the growth characteristics of a population containing that phenotype.

The existence of these two complementary interpretations to the selective value is of practical importance because the number of methods for measuring selective values is increased. The original method of measuring a selective value function, say $W_{AA}(N)$, following the definition of the selective value in equations (5), consists of determining the

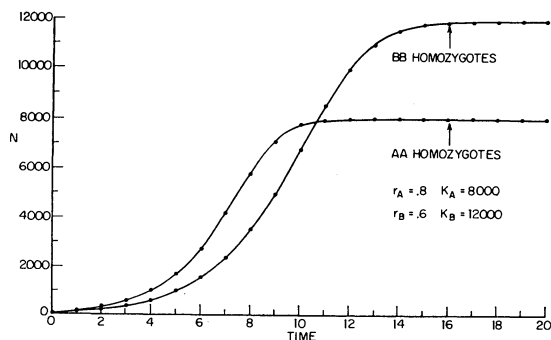


FIG. 2. Population growth for populations consisting entirely of AA or BB homozygotes.

product of the individual fertility and survival fraction, (ml_{AA}), at various densities. The best fit of this data to a line with negative slope yields an equation in N whose intercept at $N = 0$ is $(r_A + 1)$ and which equals one at $N = K_A$. An alternative method now possible is to isolate a true breeding strain with the phenotype of the AA homozygotes, and to observe the population growth curve. The best fit of the population growth data to a logistic equation provides the values for r_A and K_A .

Two complementary interpretations also exist for the heterozygote selective value. However, the method of measuring the selective value from observations on population growth may need to be somewhat contrived. One must either obtain a true breeding strain with the phenotype of the AB heterozygotes—a somewhat unlikely possibility unless perhaps convergent phenotypes are drawn from populations of different evolutionary stock—or, using actual AB heterozygotes, one must observe population growth under conditions such that he substitutes offspring of AB genotype for any offspring produced of other genotypes before differential survival is significant.

The evolutionary result of density-dependent natural selection in any environment is contingent on the fitness of the heterozygotes relative to the two homozygotes. The set of possible relations between the fitness of the heterozygote and fitness of the homozygotes generates a set of different cases of density-dependent selection, many requiring separate analysis. All the cases are listed in Table 2. The situation described by most of the cases is self-explanatory except for cases 11, 12, and 13, which all involve a heterozygote intermediate in both r and K . Referring to Fig. 3, it is clear that $W_{AB}(N)$ may pass above, through, or below the point where $W_{AA}(N)$ intersects $W_{BB}(N)$. If $W_{AB}(N)$ passes above this point of intersection, then there is an interval of population sizes in which the heterozygote

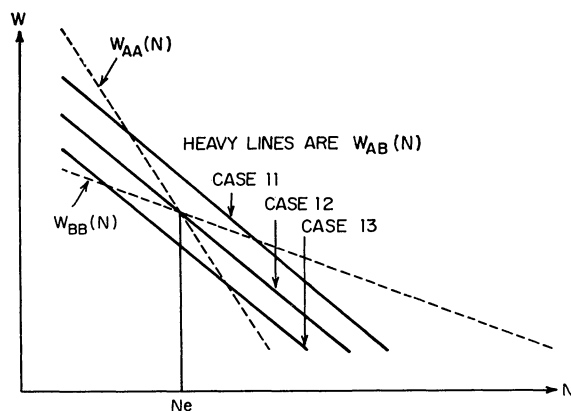


FIG. 3. Selective value of the heterozygote phenotype as a function of population density for the three cases where the heterozygote is intermediate in both r and K .

TABLE 2. Possible cases of heterozygote fitness

Case	Properties of heterozygote	Requirement on r_H and K_H	
1.....	Same as High- r homozygote	$K_H = K_A$,	$r_H = r_A$
2.....	Same as High- K homozygote	$K_H = K_B$,	$r_H = r_B$
3.....	Superior K , Superior r	$K_H > K_B$,	$r_H > r_A$
4.....	" Intermediate r	"	$r_B < r_H < r_A$
5.....	" Inferior r	"	$r_H < r_B$
6.....	Inferior K , Superior r	$K_H < K_A$,	$r_H > r_A$
7.....	" Intermediate r	"	$r_B < r_H < r_A$
8.....	" Inferior r	"	$r_H < r_B$
9.....	Intermediate K , Superior r	$K_A < K_H < K_B$,	$r_H > r_A$
10.....	" Inferior r	"	$r_H < r_B$
	Intermediate K , Intermediate r		$r_B < r_H < r_A$
11.....	Heterozygote superior at some densities	$W_{AB}(N_e) > W_{AA}(N_e)$	
12.....	Heterozygote intermediate at all densities	$W_{AB}(N_e) = W_{AA}(N_e)$	
13.....	Heterozygote inferior at some densities	$W_{AB}(N_e) < W_{AA}(N_e)$	

is superior to both homozygotes. If $W_{AB}(N)$ passes below this point of intersection, then there is an interval where the heterozygote is inferior. Only if $W_{AB}(N)$ actually passes through the point is the heterozygote intermediate in fitness at all population sizes (except N_e).

CONSTANT ENVIRONMENT

The evolutionary outcome of density-dependent natural selection is analyzed first in a constant environment. A constant environment need not be a theoretical abstraction. A natural environment is understood as constant, with respect to a particular species, if in that environment consecutive generations of the species face the same kinds of selective pressures. Since selective pressures are specified with the selective functions, a constant environment can be defined as one where the selective value functions for consecutive generations have the same functional form and whose parameters keep the same numerical values. For the model of density-dependent selective pressures, the selective values through consecutive generations must remain of the form $W(N) = (r + 1) - (r/K)N$, and the numerical values of r and K must remain the same. Thus a natural environment having much spatial and temporal heterogeneity may very well be constant for some species simply because species there receive the same kinds of selective pressures in successive generations.

Full dominance, cases 1 and 2

The case where the high- r allele, A, is fully dominant is presented in Fig. 4; where the high- K allele, B, is fully dominant in Fig. 5. The family of trajectories for both cases include certain common features:

- (1) All trajectories of the population converge eventually to an asymptotically stable equilibrium point with the population size equaling the larger of the carrying capacities and with the gene pool

fixed for the allele yielding the larger carrying capacity.

- (2) During the early generations of any trajectory, the population always shows an increase in the *proportion* of high- r phenotypes so long as the population size is less than N_e . N_e is the population size at which the selective values of the high- r and high- K phenotypes are equal. The *proportion* of high- r phenotypes decreases whenever the population size is greater than N_e .
- (3) During the early phases of any trajectory the population growth is rapid with both phenotypes increasing in *number*. A later phase is entered when the population size exceeds the smaller of the two carrying capacities, K_A . During the late phase the *number* of high- r phenotypes decreases because the selective value of that phenotype is then less than unity. The population does not grow rapidly and evolution can be interpreted as a substitution of high- K phenotypes for high- r phenotypes in a population of approximately constant size.

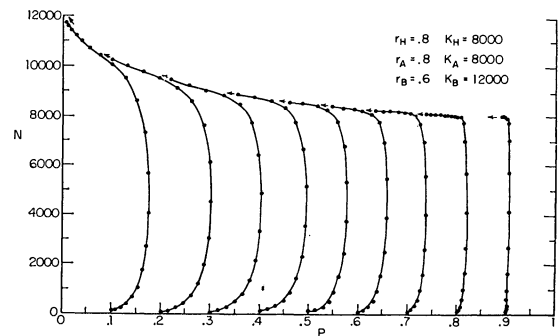


FIG. 4. Trajectories assuming the high- r allele, A, is dominant. N is the population size. The abscissa, p , is the gene frequency of A. Populations founded with 100 individuals containing various initial gene frequencies progress after each generation to the next higher dot on their trajectories; they all eventually converge at $N = 12,000$ and $p = 0$.

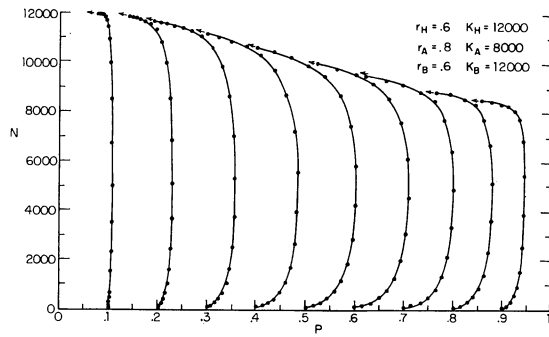


FIG. 5. Trajectories assuming the high- K allele, B, is dominant.

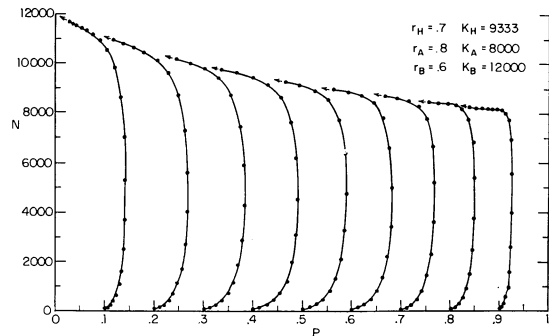


FIG. 6. Trajectories assuming the heterozygote is intermediate in fitness at all population sizes.

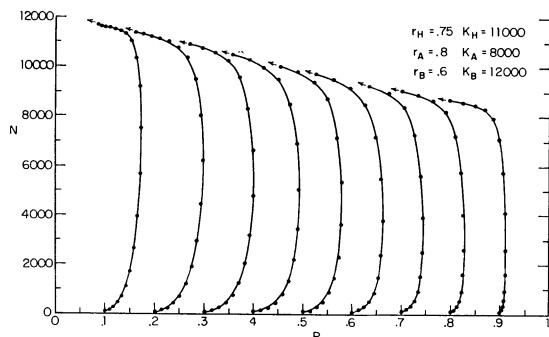


FIG. 7. Trajectories assuming the heterozygote is superior in fitness at medium population sizes but is intermediate in fitness at high and low population sizes.

The family of trajectories for both cases shows the following differences:

- (1) The plateau across the phase plane between the levels of the two carrying capacities, K_B and K_A , is at its highest elevation for the high- K allele dominant case. This result occurs because the recessive high- r allele, when present at lower frequencies, is very likely to be found in a heterozygote where it is not expressed. Consequently, the population size is large since most of the individuals possess phenotypes permitting a large carrying capacity.
- (2) The number of generations required to achieve

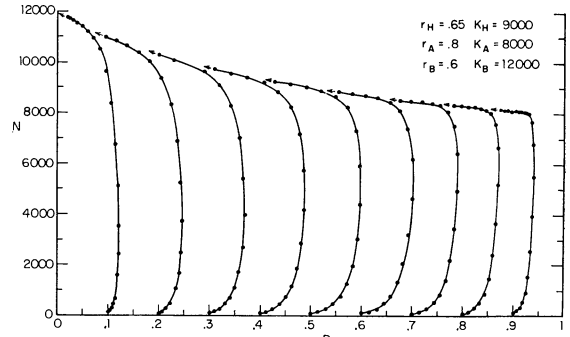


FIG. 8. Trajectories assuming the heterozygote is inferior in fitness at medium population sizes but is intermediate in fitness at high and low population sizes.

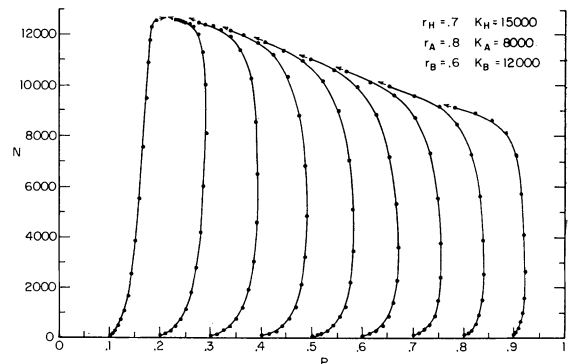


FIG. 9. Trajectories assuming the heterozygote has the largest K and is thus superior in fitness to both homozygotes at high population sizes. The trajectories converge at a stable polymorphism.

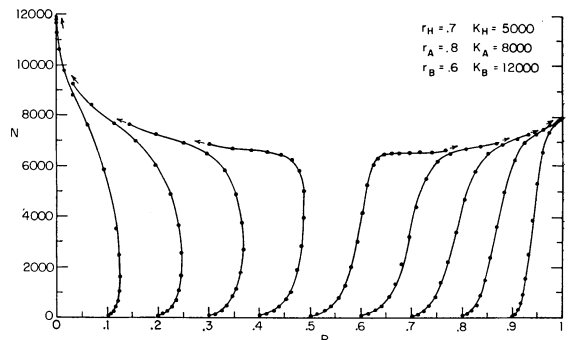


FIG. 10. Trajectories assuming the heterozygote has the lowest K and is thus inferior in fitness to both homozygotes at high population sizes. The eventual outcome of selection depends on the composition and size of the founding population.

fixation of the high- K allele, starting from intermediate gene frequencies, is highest for the high- K allele dominant case. This result is another instance of the generalization that more time is required to select against a recessive allele than a dominant allele because the presence of a recessive allele is masked in heterozygotes.

Intermediate heterozygote, cases 11, 12, and 13

The case where the heterozygote is intermediate at all population densities is presented in Fig. 6. The entire pattern of the trajectories is intermediate between the two extremes of dominance. The plateau is not at as high an elevation as in the high- K dominant case but is higher than in the high- r dominant case. The time required for a given degree of high- K allele fixation is similarly intermediate. It is important to note that the values of r_H and K_H necessary to produce this fully intermediate case are not themselves intermediate. For example, the figures presented all use $r_A = .8$, $r_B = .6$, $K_A = 8,000$, $K_B = 12,000$, but a heterozygote with $r_H = .7$ and $K_H = 10,000$ would be superior over a definite interval of population sizes, and hence would not be an instance of this fully intermediate case.

The case where the heterozygote is superior for an interval of population densities is shown in Fig. 7. The pattern differs from Fig. 6 at intermediate densities in that the trajectories lean more toward the center of the graph. For example, the segment of each trajectory between N equaling 3,000 to 7,500 and originating from p equal to .1, .2, or .3 inclines more steeply to the right in Fig. 7 than in Fig. 6, while a similar segment of each trajectory originating from p equal to .7, .8, or .9 inclines more steeply to the left. This difference does not affect the outcome of selection in a constant environment but will prove important when considering a seasonal environment later.

The case where the heterozygote is inferior over some population densities appears in Fig. 8. The pattern differs from Fig. 6 in that at intermediate densities the trajectories lean away from the center.

Heterozygote with intermediate K, cases 9 and 10

The cases where the heterozygote has an intermediate K lead to essentially the same patterns as illustrated in Fig. 4 through 8. The major differences are that with a superior r , the trajectories at low densities lean toward the center of the graph, while with an inferior r , the trajectories at low densities lean away from the center.

Heterozygote with superior K, cases 3, 4, and 5

The pattern of trajectories produced with a superior heterozygote K is shown in Fig. 9. The pattern is different from the previous. The trajectories converge at some non-zero gene frequency maintaining a stable polymorphism. The population size at the stable equilibrium point is less than K_H , the carrying capacity of a true breeding strain with the heterozygote phenotype, and greater than K_B , the larger of the homozygote carrying capacities. The equilibrium values of N and p are obtained by letting both $\Delta N = 0$ and $\Delta p = 0$ in equations (9) and (10) and

solving for the resultant N and p . The pattern leading to a stable polymorphism occurs regardless of the relative magnitude of r_H , ($r < 1$). In spite of the special pattern of trajectories associated with a superior K heterozygote, the result of natural selection is, as before, the maintenance of the phenotype with the highest K in as high a proportion as possible in the population.

Heterozygote with inferior K, cases 6, 7, and 8

The pattern of trajectories for a heterozygote with an inferior K is presented in Fig. 10. This pattern also is quite different from any of the patterns previously discussed. The trajectories lead to fixation of either the high- r or high- K allele depending upon the initial gene frequency. This pattern occurs regardless of the relative magnitude of r_H , ($r < 1$).

GRAPHICAL ANALYSIS OF SEASONAL ENVIRONMENT

Many, perhaps most, species of organisms in the world live in an environment with intense seasonal variability. Do the conclusions derived in a constant environment still apply for a seasonal environment? Records of environmental variation obtained with thermometers, etc., are not enough to entail that the previous constant environment analysis is inappropriate for any given species. The necessary and sufficient condition that the constant environment treatment be inappropriate is that members of consecutive generations in the given species face selective pressures specified by different equations, or by the same equations but with changed parameter values. Two examples: in one generation the selective values show negative linear-density dependence and in the next generation follow some other equation; in one generation the r 's and K 's take one set of values and in the next generation an alternate set of values. Clearly this condition disallowing a constant environment analysis bears more heavily in some groups of organisms than in others. One can imagine the same habitat being a constant environment to a vertebrate while being a seasonal environment to an insect.

We will assume, for simplicity, a particular kind of seasonal environment. A very distinct growing season and season of dormancy occur in this environment. Throughout the growing season the density-dependent selective values obey equations (11) and the population changes just as if it were in a constant environment. However, during the dormancy season the population does not function well, with the result that at the end of each dormancy season the population is left with only N^0 individuals possessing the gene frequency which occurred at the end of the preceding growing season. Thus, each growing season starts with a founding population of N^0 individuals with the gene pool left from the end of the preceding growing season. Considerable atten-

tion will be focused on N^0 which, for a growing season of some given number of generations, serves as a good measure of the harshness of the environment. The seasonal environment described above lends itself very readily to analysis by a graphical technique.

Full dominance

Consider first the case where the high- r allele is dominant. The trajectories appear in Fig. 4. One must specify two parameters of the seasonal environment in advance: N^0 , the size of the founding population at the beginning of each growing season, and L , the length of the growing season in generations. Initially, let L be two generations. Then consider a harsh environment that causes N^0 to be 2,500. Draw a horizontal line across the phase plane at the level of N equal to 2,500. Next estimate the vertical distance the population will have moved after two generations starting at N equal to 2,500. This new vertical distance is about at N equal to 5,500. Draw a horizontal line at this level. One can now estimate the trajectory of the population through successive growing seasons. Suppose the population starts out originally at N equal to 2,500 and p equal to .1; then at the end of the first season N is about 5,500 and p is about .12. The beginning of the next growing season is entered with p equal to .12 and N equal to 2,500, and so on. The result after successive seasons is that p gets progressively larger until eventually the high- r allele is fixed. The same result will clearly occur if the environment is harsher making N^0 lower, assuming L is still two generations.

Now consider a milder environment with N^0 equal to 3,900 but with L still two generations. For this environment successive growing seasons lead eventually to fixation of the high- K allele. Clearly the same result will occur in milder environments with growing seasons of two generations in length. Finally, consider an environment with N^0 equal to 2,500 again but with L set at four generations. For this environment evolution in successive growing seasons leads to fixation of the high- K allele.

Thus, the outcome of natural selection in a harsh seasonal environment may be entirely opposite to that in the constant environment because the high- r allele can be fixed instead of the high- K allele. However, the long-term intensity of natural selection in favor of the high- r allele is diminished as the dormancy season becomes milder and/or as the growing season becomes longer. Indeed, if the dormancy season is mild enough or the growing season long enough, the high- K allele will be fixed instead of the high- r allele. These results are also true for the case where the high- K allele is dominant, as presented in Fig. 5. The limitation of the graphical technique is clearly one of accuracy. It is impossible to determine

graphically the degree of harshness marking the transition between selection for the high- r allele instead of the high- K allele. This question and others of a quantitative sort about the seasonal environment are deferred until the next section.

Intermediate heterozygote

The case where the heterozygote shows superiority through an interval of intermediate densities has some interesting properties in a seasonal environment. By applying the graphical method to Fig. 7, one can verify that with reasonably short growing seasons, about two to four generations, three results can occur depending upon the harshness of the environment. If the environment is very harsh, with N^0 in the range of 1,500 and below, then the high- r allele is either fixed, or exists in very high frequency. If the environment is very mild, with N^0 in the range of 6,000 and above, then the high- K allele is either fixed or exists in very high frequency. But if the environment is of intermediate harshness with N^0 in the range of 1,500 to 6,000, then a significant stable polymorphism results. A more accurate analysis is presented in the next section.

The case where the heterozygote shows inferiority through an interval of intermediate densities can also show three results with reasonably short growing seasons depending upon the harshness of the environment. A very harsh environment results in fixation of the high- r allele, a mild environment in fixation of the high- K allele, while an environment of intermediate harshness results either in fixation of the high- r allele or the high- K allele, depending on the original gene frequency in the population.

The case where the heterozygote is of intermediate fitness at all densities shows the same results as the cases with full dominance previously discussed.

Additional cases

The additional cases involve a heterozygote superior in r or inferior in r . As might be expected, in a harsh environment with a short growing season, evolution with a heterozygote superior in r leads to a stable polymorphism, while with a heterozygote inferior in r , evolution leads either to fixation of the high- r or of the high- K allele, depending upon the original gene pool frequency. These evolutionary results for a harsh environment with a short growing season are independent of the value of the heterozygote K .

Let us agree to call selection which results in evolution of high- K phenotypes at the expense of a low r as " K -selection," and that which results in evolution of high- r phenotypes at the expense of a low K as " r -selection." The model, as discussed so far, clearly indicates that K -selection occurs in constant and mild seasonal environments and r -selection in

harsh seasonal environments with short growing seasons. However, K -selection and r -selection do not exhaust the possible outcomes, for we have already seen one case, with a phenotypically intermediate heterozygote, where density-dependent selection in a seasonal environment of medium harshness results in polymorphism, and more such cases are discussed in the next section.

Another kind of environment leading to r -selection, discussed by Gadgil and Bossert (1970), presumes some continuously operating cause of mortality extrinsic to the population itself. This mortality is assumed to maintain the population density at a low enough level so that the selective value of the high- K phenotype is still less than that of the high- r phenotype. Such an environment leads to the fixation of the high- r gene. This environment would be appropriate, for example, to weeds in a field where mowing regularly holds the plant density to a low value.

MacArthur and Wilson (1967) proposed that there should exist a latitudinal gradient in r -selection and K -selection; that proportionally more species in tropical latitudes face K -selection instead of r -selection than do species of temperate latitudes. Scientists working in the tropics often doubt the truth of this hypothesis because the evident seasonal variability in tropical habitats appears to disqualify these habitats from being the uniformly benign environments where K -selection was supposed to occur. But, as we have seen, seasonal variability, per se, does not rule out K -selection. It is the magnitude of harshness that is critical. And it seems true that the harshness of the seasonal variability in most (not all) tropical habitats is not as large as in comparable temperate habitats. So the hypothesis that there is a latitudinal gradient in r -selection and K -selection should be taken seriously. However, the gradient should only occur in certain groups, groups to whom the seasonal variability appears coarse grained. Long-lived vertebrates are probably facing K -selection all over the world, but it may be otherwise with some insects and other invertebrates with short life span.

The hypothesis of a latitudinal gradient in r -selection and K -selection is testable. We know, for example, that growth of genetically homogeneous populations of *Drosophila melanogaster* follows the logistic curve (Buzzati-Traverso 1955, p. 178). It would be interesting to see whether the r 's of different *Drosophila* populations are positively correlated with latitude and the K 's negatively correlated.

MATHEMATICAL ANALYSIS OF SEASONAL ENVIRONMENT

We now turn to the mathematical analysis of evolution in seasonal environments.

The distinguishing feature of the seasonal environ-

ment is that consecutive generations are expected to face different regimes of selective pressures. The eventual outcome of evolution can only be anticipated by knowing the long-term fitness of the various phenotypes. The long-term fitness must refer to the performance of a phenotype over the number, L , of generations required for the entire expected regime of selective pressures to be realized. This number, L , is called the period of the environment, and the seasonal environment is seen as a continuing repetition of a sequence of L regimes of selective pressures. One approach to the analysis of the seasonal environment is to construct a measure of the long-term fitness from the set of single-generation selective values representing each of the expected regimes of natural selection.

Suppose initially that the period of the environment is two generations. The two regimes of selective pressures that occur are represented by two sets of selective value functions, e.g., $W_{AA,0}(N, p)$ and $W_{AA,1}(N, p)$. The numerical subscript refers to a position in the sequence of occurrence of selective regimes; thus within any period $W_{AA,0}(N, p)$ acts for the initial generation and $W_{AA,1}(N, p)$ for the next. Let p^0 be the frequency of the A allele at the beginning of a period, p^1 be the frequency in the middle, and p^2 the final frequency. (Remember, superscripts in italics are not exponents; they serve to identify the generation.) Then from the definition of the selective values (5) and the considerations expressed back in Table 1,

$$\begin{aligned} N_{AA}^2 &= W_{AA,1}(N^1, p^1) (p^1)^2 N^1 \\ N_{AB}^2 &= W_{AB,1}(N^1, p^1) 2p^1 q^1 N^1 \\ N_{BB}^2 &= W_{BB,1}(N^1, p^1) (q^1)^2 N^1 \end{aligned} \quad (13)$$

However [see equations (6) and (8)],

$$\begin{aligned} p^1 &= \frac{p^0 W_{AA,0}(N^0, p^0) + q^0 W_{AB,0}(N^0, p^0)}{\bar{W}_0(N^0, p^0)} p^0 \\ q^1 &= \frac{q^0 W_{BB,0}(N^0, p^0) + p^0 W_{AB,0}(N^0, p^0)}{\bar{W}_0(N^0, p^0)} q^0 \\ N^1 &= \bar{W}_0(N^0, p^0) N^0 \end{aligned} \quad (14)$$

Substituting (14) into (13) yields expressions for the genotypic numbers at the end of an environmental period in terms of p and N at the beginning of the period. These expressions are of the form

$$\begin{aligned} N_{AA}^2 &= A_{AA,2}(N^0, p^0) (p^0)^2 N^0 \\ N_{AB}^2 &= A_{AB,2}(N^0, p^0) 2p^0 q^0 N^0 \\ N_{BB}^2 &= A_{BB,2}(N^0, p^0) (q^0)^2 N^0 \end{aligned} \quad (15)$$

where

$$A_{AA,2}(N^0, p^0) = W_{AA,1}(N^1, p^1) \times \frac{[p^0 W_{AA,0}(N^0, p^0) + q^0 W_{AB,0}(N^0, p^0)]^2}{\bar{W}_0(N^0, p^0)}$$

$$A_{AB,2}(N^0, p^0) = W_{AB,1}(N^1, p^1) \times \frac{[p^0 W_{AA,0}(N^0, p^0) + q^0 W_{AB,0}(N^0, p^0)] \times [q^0 W_{BB,0}(N^0, p^0) + p^0 W_{AB,0}(N^0, p^0)]}{\bar{W}_0(N^0, p^0)} \quad (16)$$

$$A_{BB,2}(N^0, p^0) = W_{BB,1}(N^1, p^1) \times \frac{[q^0 W_{BB,0}(N^0, p^0) + p^0 W_{AB,0}(N^0, p^0)]^2}{\bar{W}_0(N^0, p^0)}$$

The functions defined in (16) are the net selective values for a seasonal environment with a period of two generations. The net selective value (n.s.v.) functions appear in equations for the change in N and p over a seasonal cycle that are exactly parallel to the equations for ΔN and Δp over one generation. By arguments like those used in deriving (6) and (8) one obtains

$$p^2 = \frac{p^0 A_{AA,2}(N^0, p^0) + q^0 A_{AB,2}(N^0, p^0)}{\bar{A}_2(N^0, p^0)} p^0 \quad (17)$$

$$N^2 = \bar{A}_2(N^0, p^0) N^0 \quad (18)$$

Equations (17) and (18) are exactly the same in form as (6) and (8) and, indeed, the similarity is fundamentally due to the parallel form of (13) and (15).

The n.s.v. functions derived here differ systematically from the "coarse grained adaptive function" proposed by Levins (1969). Levins' adaptive function for L generations is a product of L selective values, while equations (16) indicate that with L equaling two, an n.s.v. is the product of three selective values. This discrepancy increases as L becomes larger. It is not clear whether conclusions obtained as the condition for maximizing Levins' adaptive function are compatible with those obtained using the n.s.v.'s in equations (17) and (18).

An interesting property of the n.s.v. functions is that even with full phenotypic dominance the n.s.v. of the heterozygote does not equal that of the dominant homozygote. Full phenotypic dominance entails, assuming A is dominant, that $W_{AA,0}(N, p) = W_{AB,0}(N, p)$ and that $W_{AA,1}(N, p) = W_{AB,1}(N, p)$. Substituting these equalities into (16) indicates that $A_{AA,2}(N, p) \neq A_{AB,2}(N, p)$. These n.s.v.'s are not equal because the expected offspring from heterozygotes differ considerably from those of homozygotes even though their phenotypes otherwise are the same. The n.s.v., in covering more than one generation, takes accounts of this fundamental difference between heterozygotes and homozygotes.

The n.s.v. functions for a seasonal environment with an arbitrary period, say $j+1$, can be generated with the following recursive rules:

$$A_{AA,j+1}(N^0, p^0) = W_{AA,j}(N^j, p^j) \times \frac{[p^0 A_{AA,j}(N^0, p^0) + q^0 A_{AB,j}(N^0, p^0)]^2}{\bar{A}_j(N^0, p^0)}$$

$$A_{AB,j+1}(N^0, p^0) = W_{AB,j}(N^j, p^j) \times \frac{[p^0 A_{AA,j}(N^0, p^0) + q^0 A_{AB,j}(N^0, p^0)] \times [q^0 A_{BB,j}(N^0, p^0) + p^0 A_{AB,j}(N^0, p^0)]}{\bar{A}_j(N^0, p^0)} \quad (19)$$

$$A_{BB,j+1}(N^0, p^0) = W_{BB,j}(N^j, p^j) \times \frac{[q^0 A_{BB,j}(N^0, p^0) + p^0 A_{AB,j}(N^0, p^0)]^2}{\bar{A}_j(N^0, p^0)}$$

The N^j , and p^j appearing in (19) are obtained after j iterations with (14) on N^0 and p^0 using the appropriate selective values in each iteration. An n.s.v. for an environment with a period of 1 generation is simply the usual selective value itself, so, $A_{AA,1}(N, p) = W_{AA,0}(N, p)$, etc.

The n.s.v.'s can be used to solve for a kind of steady state in a seasonal environment, obtained when $p^2 = p^0$ and $N^2 = N^0$ in equations (17) and (18). The defining characteristic of this steady state is that N and p are the same at the start of consecutive seasons, although they change values within a season. Solving for this steady state is especially easy in the sort of seasonal environment we have been considering, where each growing season begins with a population size N^0 . In this environment N^0 ceases to be a variable and is instead a parameter, and (18) is then suppressed. The condition for the (nontrivial) steady state from (17) is that $(pA_{AA} + qA_{AB})/\bar{A} = 1$. However $(pA_{AA} + qA_{AB} - \bar{A})$ can be rewritten as $q[p(A_{AA} - A_{AB}) - q(A_{BB} - A_{AB})]$. So an equilibrium is achieved at $\hat{p}$ obtained from $\hat{p}^0[A_{AA}(N^0, p^0) - A_{AB}(N^0, p^0)] - \hat{q}^0[A_{BB}(N^0, p^0) - A_{AB}(N^0, p^0)] = 0 \quad (20)$

Equation (20) brings out the conclusion that a steady state will occur in a seasonal environment at some frequency only if the heterozygote n.s.v. at that frequency is superior or inferior to both homozygote n.s.v.'s there.

The conditions for stability of the equilibrium in the seasonal environment are not so readily established. Denoting the RHS of (17) by $f(p^0)$, the stability properties at equilibrium are determined from $f'(\hat{p}^0)$, i.e., the derivative of $f(p^0)$ evaluated at equilibrium. If $f'(\hat{p}^0)$ is in $(0, 1)$ perturbations from equilibrium return smoothly, in $(-1, 0)$ the return is with damped oscillations, and if $|f'(\hat{p}^0)| > 1$ the equilibrium is unstable. The condition for stability requires, when the selective values do not depend on p , that the heterozygote must be superior, but the situation is more complicated when the selective values do depend on p . We consider later an equi-

librium which is stable and where the heterozygote n.s.v. is inferior to both homozygote n.s.v.'s. Two kinds of cases are considered next, one where there is full dominance, the other where the heterozygote is a phenotypic intermediate.

Full dominance

We now show that when the high- r allele, A , is dominant there are certain levels of environmental harshness, measured by N^0 , in which the heterozygote n.s.v. is inferior to both homozygote n.s.v.'s. Recall that N_e is the population density at which $W_{AA} = W_{BB}$. If $N^0 < N_e$ and $N^1 > N_e$, then the dominant phenotype, AA or AB , is fittest in the initial generation and the recessive phenotype, BB , is fittest in the next generation of the growing season. When A is dominant

$$A_{AA}(N^0, p^0) - A_{AB}(N^0, p^0) = \{W_{AA}(N^0, p^0) - [q^0 W_{BB}(N^0, p^0) + p^0 W_{AA}(N^0, p^0)]\} \times W_{AA}(N^1, p^1) p^1 / p^0$$

$$A_{BB}(N^0, p^0) - A_{AB}(N^0, p^0) = \{W_{BB}(N^1, p^1) - [q^0 W_{BB}(N^0, p^0) + p^0 W_{AA}(N^0, p^0)]\} \times W_{AA}(N^1, p^1) W_{AA}(N^0, p^0) / q^1 / q^0$$

When $N^0 < N_e$ and $N^1 > N_e$, $A_{AA} - A_{AB}$ is positive for all p^0 making the heterozygote n.s.v. always inferior to the AA homozygote n.s.v. If p^0 is close to 1, $A_{BB} - A_{AB}$ is positive if

$$[W_{BB}(N^1, p^1) - W_{AA}(N^1, p^1)] W_{AA}(N^0, p^0)$$

is positive, which is always true when $N^1 > N_e$. If p^0 is close to zero, $A_{BB} - A_{AB}$ is positive if

$$W_{BB}(N^1, p^1) W_{BB}(N^0, p^0) - W_{AA}(N^1, p^1) W_{AA}(N^0, p^0)$$

is positive, and this condition is true when, so to speak, N^0 is not as far below N_e as N^1 is above. So if N^0 is low, the heterozygote n.s.v. is inferior to the BB homozygote n.s.v. only at high p^0 , but as N^0 approaches N_e the heterozygote is inferior for an increasingly wider interval of p^0 . These results satisfy a necessary condition for the existence of polymorphism caused by the seasonal environment.

By similar reasoning it can be shown that when the high- K allele, B , is dominant, the heterozygote n.s.v. is superior to both homozygote n.s.v.'s when $N^0 < N_e$ and $N^1 > N_e$, and again a necessary condition for the existence of polymorphism is satisfied.

To determine if such polymorphism actually exists (20) was solved numerically for a range of N^0 and the derivative of (17) examined at any equilibrium obtained to see if the equilibrium was stable. The results, presented in Fig. 11, indicate that there is a stable polymorphism over a small range of environmental harshness for both cases of full dominance. The degree of stability, as measured by the rate of return to equilibrium after a small perturbation, is low. The deviation from equilibrium after t growing seasons, $d(t)$, given an initial small displacement d_0 , is given by $d(t) = [f'(\hat{p}^0)]^t d_0$ where, as before, $f'(\hat{p}^0)$ is the derivative of (17) evaluated at equilibrium. In the numerical cases presented in Fig. 11, $f'(\hat{p}^0)$ at its stablest was .9989. Fig. 11 shows that where the high- r allele A is recessive its equilibrium frequency, $\hat{p}^0$, is higher for a given N^0 than where A is dominant. Hence for a given harshness, N^0 , the observable difference in phenotypic proportions produced by the two genetic mechanisms may be small.

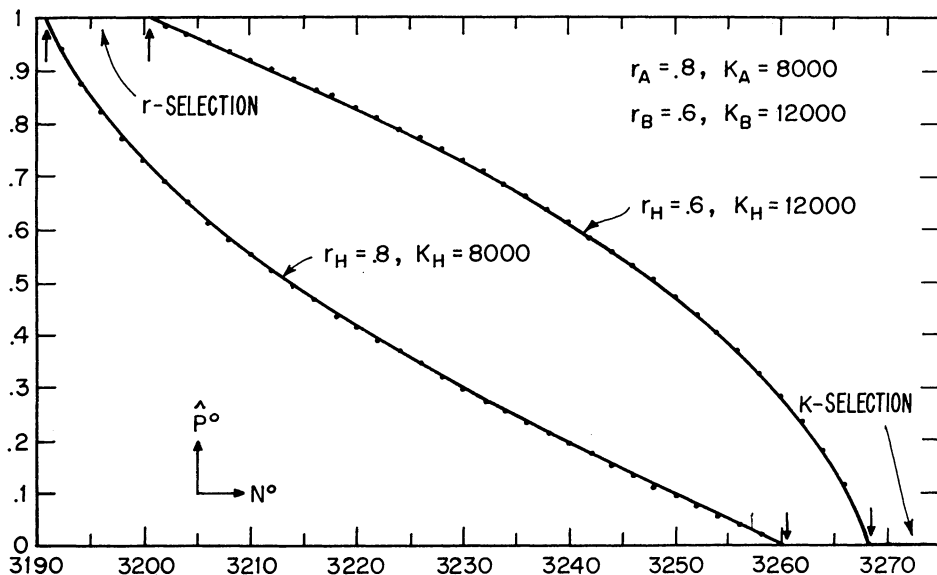


FIG. 11. Polymorphism frequency, $\hat{p}^0$, as a function of environmental harshness, assuming full dominance. Growing season is two generations long. Arrows indicate the end points of the harshness intervals.

The existence of this polymorphism caused by a seasonal environment can be explained by, and could perhaps have been deduced from, an extension of the work by Haldane and Jayakar (1963) on selection in changing environments. They consider a coarse-grained environment where the fitness of the phenotypes varies among different generations, but the fitness for every generation is a constant, i.e., neither frequency nor density dependent. However, in the model considered here, the fitness in the second generation, e.g., $W_{AA}(N^1)$, is a function of the frequency at the beginning of the growing season because $N^1 = \bar{W}(N^0, p^0)N^0$. (Remember, N^0 is a parameter.) So Haldane and Jayakar's results must be extended to frequency-dependent fitnesses, an easy adaptation as it turns out.

Suppose A, the high- r gene is dominant. Let the ratio of p^i to q^i be $u_i = p^i/q^i$. Let F_i be the relative fitness of the recessive in generation i : $F_i(N^i) = W_{BB}(N^i)/W_{AA}(N^i)$. When A is dominant Haldane (1932, p. 181) showed that $u_{i+1} = u_i(u_i + 1)/(u_i + F_i)$. By simple algebraic manipulation Haldane and Jayakar (1963) arrive at two particularly expressive formulas for the difference in u after n generations:

$$u_n - u_0 = n - \sum_{i=0}^{n-1} F_i + \sum_{i=0}^{n-1} \frac{F_i(F_i - 1)}{u_i + F_i} \quad (21a)$$

$$\log u_n - \log u_0 = - \sum_{i=0}^{n-1} \log F_i - \sum_{i=0}^{n-1} \log \left[1 + \frac{(1 - F_i)u_i}{F_i(1 + u_i)} \right] \quad (21b)$$

We consider now what happens to trajectories over n generations starting from various p^0 . Suppose p^0 is near zero, so that u_0 and all subsequent u_i are near zero. If p^0 is small enough, the second term in (21b) can be ignored and then p^n will be larger than p^0 if $\sum \log F_i < 0$ where each F_i is evaluated at the N^i along the trajectory starting at N^0 and $p^0 \approx 0$. $\sum \log F_i < 0$ is equivalent to $\prod F_i < 1$. Thus trajectories starting from very small p^0 will lead to higher p^n provided the geometric mean of the relative fitnesses along the trajectories is less than one. Suppose next that p^0 is near one so that u_0 and subsequent u_i are near infinity. If p^0 is close enough to one, then the second term in (21a) can be ignored, and p^n will be less than p_0 if $\sum F_i > n$ where the F_i are evaluated at the N^i along the trajectory starting from N^0 and $p^0 \approx 1$. $\sum F_i > n$ is equivalent to $(1/n)\sum F_i > 1$. Thus trajectories starting from large p^0 lead to lower p^n provided the arithmetic mean of the relative fitnesses along the trajectories is greater than one. The difference between this discussion and Haldane and Jayakar's is simply that here we must be careful to take the means of the F_i along certain

trajectories, while with Haldane and Jayakar's constant fitnesses, the means of the F_i are the same along all trajectories.

The results about the means of the F_i are easily applied to our two-generation seasonal environment. We consider what happens as an environment becomes progressively harsher. First, in a mild environment the recessive high- K allele B is fixed, so the trajectories from any p^0 lead to a lower p^n . But as the environment becomes harsher, at some N_u^0 , the trajectories from low p^0 lead to a higher p^n . This N_u^0 is the upper limit of the interval of environmental harshness where the seasonal environment causes polymorphism. N_u^0 is found from the condition on the geometric mean at low p^0 :

$$(F_0 F_1)^{\frac{1}{2}} = \left[\frac{W_{BB}(N_u^0)}{W_{AA}(N_u^0)} \times \frac{W_{BB}[W_{BB}(N_u^0)N_u^0]}{W_{AA}[W_{BB}(N_u^0)N_u^0]} \right]^{\frac{1}{2}} = 1 \quad (22a)$$

The lower end of this interval occurs when trajectories at high p^0 cease to lead to lower p^n . The lower end, N_l^0 , is found from the condition on the arithmetic mean at high p^0 :

$$\frac{1}{2}(F_0 + F_1) = \frac{1}{2} \left[\frac{W_{BB}(N_l^0)}{W_{AA}(N_l^0)} + \frac{W_{BB}[W_{AA}(N_l^0)N_l^0]}{W_{AA}[W_{AA}(N_l^0)N_l^0]} \right] = 1 \quad (22b)$$

In harsh environments, below N_l^0 , the dominant high- r allele, A, is fixed. The equations for N_u^0 and N_l^0 become cubic equations and the arrows in Fig. 11 for the interval end points indicate the solutions of these equations.

Haldane and Jayakar (1963) pointed out, using their constant fitnesses and full dominance, that a changing environment could never lead to an unstable equilibrium. An unstable equilibrium occurs if the geometric mean of the relative fitness is greater than one at low p^0 and the arithmetic mean less than one at high p^0 . But since the geometric mean of the fitnesses in a changing environment is necessarily less than the arithmetic mean, the condition for an unstable equilibrium is impossible with constant fitnesses. An unstable equilibrium is possible, however, with frequency-dependent fitnesses. We now show that an unstable equilibrium caused by the seasonal environment cannot occur in the model considered here.

Suppose again the high- r allele, A, to be dominant. To obtain any equilibrium we had to assume that $N^0 < N_e$ and $N^1 > N_e$. Then $W_{BB}(N^0)/W_{AA}(N^0) < 1$ and $W_{BB}(N^1)/W_{AA}(N^1) > 1$. By the instability condition the geometric mean at low p^0 is greater than one, and, since the arithmetic mean at high p^0 is less than one, then the geometric mean at high p^0 is also

less than one. So a necessary condition of the unstable equilibrium is

$$\frac{W_{BB}(N^0)}{W_{AA}(N^0)} \times \frac{W_{BB} [W_{BB}(N^0)N^0]}{W_{AA} [W_{BB}(N^0)N^0]} > 1$$

and

$$\frac{W_{BB}(N^0)}{W_{AA}(N^0)} \times \frac{W_{BB} [W_{AA}(N^0)N^0]}{W_{AA} [W_{AA}(N^0)N^0]} < 1$$

therefore

$$\frac{W_{BB} [W_{BB}(N^0)N^0]}{W_{AA} [W_{BB}(N^0)N^0]} > \frac{W_{BB} [W_{AA}(N^0)N^0]}{W_{AA} [W_{AA}(N^0)N^0]}$$

which, however, is only true if $W_{BB}(N^0)N^0 > W_{AA}(N^0)N^0$ since B is the high- K allele. But this last inequality entails that B is the high- r allele, which contradicts the premise. So for a two-generation seasonal environment to cause an unstable equilibrium with full dominance requires a more elaborate scheme of frequency-dependent selection.

The case where the high- K allele is dominant, rather than recessive, can most easily be treated by letting p still stand for the dominant allele frequency, now B rather than A. The upper limit, N_u^0 , to the harshness interval is found from

$$\frac{1}{2} \left[\frac{W_{AA}(N_u^0)}{W_{BB}(N_u^0)} + \frac{W_{AA} [W_{BB}(N_u^0)N_u^0]}{W_{BB} [W_{BB}(N_u^0)N_u^0]} \right] = 1 \quad (22c)$$

and the lower limit, N_l^0 , from

$$\left[\frac{W_{AA}(N_l^0)}{W_{BB}(N_l^0)} \times \frac{W_{AA} (W_{AA}(N_l^0) N_l^0)}{W_{BB} (W_{AA}(N_l^0) N_l^0)} \right]^{\frac{1}{2}} = 1 \quad (22d)$$

It can also be shown that the seasonal environment cannot cause an unstable equilibrium with this model.

Phenotypically intermediate heterozygote

The earlier graphical analysis of evolution in the seasonal environment showed that if a phenotypically intermediate heterozygote was superior to both homozygotes at moderate population densities, then a moderately harsh environment would cause polymorphism. Solving (20) for various N^0 revealed the nature of this polymorphism as a function of environmental harshness. The results appear in Fig. 12. Clearly the interval of environmental harshness in which the polymorphism occurs is considerably wider than the interval where there is full dominance as in Fig. 11. Also, the equilibria are somewhat more stable, with $f'(\hat{p}^0)$ at its stablest being .9849.

It should be mentioned that the phenotypically intermediate heterozygote whose trajectories appear in Fig. 6 also had a small interval of harshness causing stable polymorphism. And, as indicated by the earlier graphical analysis, an intermediate heterozygote inferior at moderate densities has an interval of harshness where an unstable equilibrium occurs.

We see that a variety of genetic mechanisms gives

rise to a transition zone between K -selection and r -selection within which a stable polymorphism is maintained. However, the mechanism involving a phenotypically intermediate heterozygote superior at moderate population densities shows polymorphism under the widest range of environmental harshness and genetic properties and hence, of those considered, is the mechanism most likely to occur. Perhaps such polymorphism accounts in part for the large amount of heterozygosity observed in many natural populations, e.g., the study of Lewontin and Hubby (1966).

ECOLOGICAL ENERGETICS AND r AND K

In this section we explore the phenotypic properties caused to evolve under density-dependent selection. We are not interested in species-specific characters so much as those characteristics applicable to any organism that indicate how an organism participates in the overall energy flow through his population.

We refer to the same population as before but assume that the locus is fixed for either the A or B allele. Therefore the population shows logistic growth as in equation (12). As before, the population is censused at the time of interbreeding.

The strategy of this section is to derive, completely within the realm of ecological energetics, an equation describing population growth. This new equation can then be compared with the logistic equation (12) which was derived from independent premises. The comparison yields expressions for r and K in terms of concepts and measurements belonging to the field of ecological energetics.

Some basic symbols, following Lindeman's (1942) notation are defined:

- Λ^i = total standing crop energy of the population at the time of interbreeding. The adults at this time are presumed to have ripe gonads. The caloric content of males and females is identical. As before, the superscript denotes the generation.
- λ_{in}^i = total amount of energy received by the population between the interbreeding time of the i th generation and the interbreeding time of the $(i+1)$ th generation.
- λ_{out}^i = total amount of energy dissipated by the population between the interbreeding time of the i th generation and the interbreeding time of the $(i+1)$ th generation.

The fundamental equation in the theory of ecological energetics is the energy balance equation, which follows from the conservation of energy law:

$$\Delta \Lambda^i = \Lambda^{i+1} - \Lambda^i = \lambda_{in}^i - \lambda_{out}^i \quad (23)$$

For the energy balance equation to be of much biological relevance, input and output terms need to be

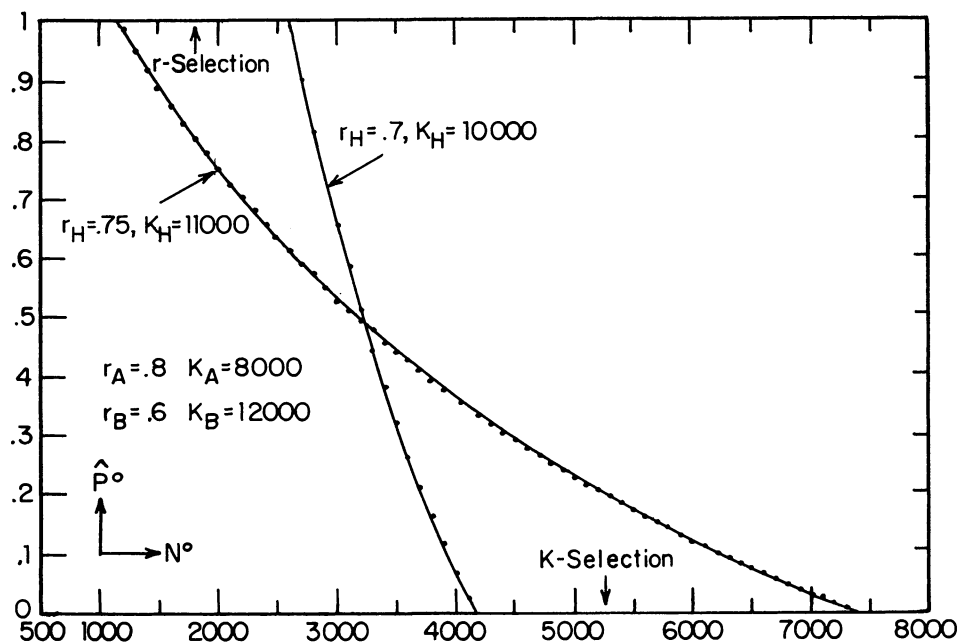


FIG. 12. Polymorphism frequency as a function of environmental harshness, assuming the heterozygote is superior in fitness at medium population sizes and intermediate in fitness at high and low population sizes. Growing season is two generations long.

expanded to indicate in more detail where energy is going and where it is coming from.

The energy dissipated by the population can be subdivided into three categories. The first category is the energy of decomposition of the bodies of the adults who have finished interbreeding. Let E be the average caloric content of an adult with ripe gonads, e be the average caloric content of a zygote, and $2m$ be the average number of zygotes produced in a mating. The average caloric content of an adult after mating can be put as $E - me$. Then the total energy dissipated from the bodies of the adults after mating is $(E - me)N^i$. Second, the organisms which survive from the zygote stage to the time of interbreeding dissipate energy throughout their lives in respiration. Let H be the average energy respired by an organism surviving from zygote to the time of interbreeding, and f be the fraction of the zygotes which survive to the time of interbreeding. Then the total energy dissipated in this category is $HfmN^i$ assuming there are $N^i/2$ matings. Third, the bodies of the organisms which do not survive from the zygote stage to adulthood dissipate an energy of decomposition. The average energy dissipated per organism in this way can be put as e , the average caloric content of a zygote, on the understanding that any energy received by such organisms before death will not be credited in the energy input computations. The total energy in this category is then $e(1 - f)mN^i$. All the above considerations are summarized as

$$\lambda_{\text{out}}^i = [E - me + Hfm + e(1 - f)m]N^i \quad (24)$$

The energy received by the population is more simply expressed. Let R be the average total energy incident upon or in other ways received by an individual from the zygotic stage until the time of interbreeding, and t be the fraction of the energy received which is actually transported across the cell membranes and incorporated into the organism. Then,

$$\lambda_{\text{in}}^i = RtfmN^i \quad (25)$$

All the previous considerations can be consolidated with the definition of the dissipation factor, D , and the input factor, I :

$$\begin{aligned} D &= E - me + Hfm + e(1 - f)m \\ I &= Rtfm \end{aligned} \quad (26)$$

The actual values of D and I are empirically determined according to formula (26). D and I both have dimensions of energy/organism. The energy balance equation is now

$$\Delta\lambda^i = IN^i - DN^i \quad (27)$$

Thus I and D are the individual's average contribution, respectively, to the energy input and to the energy dissipation of the population.

At this point we incorporate a specific sort of density dependence into the model. Assume that D and I can be written as

$$\begin{aligned} D(N) &= D_0(1 - dN) \\ I(N) &= I_0(1 - cN) \end{aligned} \quad (28)$$

where D_0 and I_0 are the energy dissipation and input factors in the absence of crowding, and d and c are crowding coefficients. Substituting (28) into (27) yields a density-dependent energy balance equation,

$$\Delta\Lambda^i = I_0(1 - cN^i)N^i - D_0(1 - dN^i)N^i \quad (29)$$

Equation (29) is the analogue in ecological energetics of the logistic equation of population dynamics:

$$\Delta N^i = rN^i \frac{(K - N^i)}{K} \quad (30)$$

The energy balance equation and the logistic equation are now to be compared. Recall that E refers to the average caloric content of an adult about to interbreed, and therefore

$$\Lambda^i = EN^i \quad (31)$$

If E itself does not change from generation to generation, then

$$\Delta\Lambda^i = E\Delta N^i \quad (32)$$

According to equation (32) the logistic equation, (30), can be multiplied by E and set equal to the energy balance equation, (29). Upon equating coefficients of N and of N^2 , one discovers

$$r = (1/E)(I_0 - D_0) \quad (33)$$

$$K = \frac{I_0 - D_0}{cI_0 - dD_0}$$

Equations (33) are expressions for the intrinsic rate of increase and the carrying capacity in terms of quantities customary to ecological energetics.

The expressions for r and K are to some extent interpretable in terms of the concepts of productivity and efficiency. One may assume for most biological systems that the energy input is either subsequently stored in the organization of biomass or variously dissipated. If the productivity of such a system is defined as the amount of energy put into biomass per unit time, then the productivity is clearly equal to the input minus the dissipation. If the efficiency of the system is defined as the amount of energy put into biomass per unit input, then the efficiency equals the difference between the input and dissipation divided by the input.

Inspection of equations (33) reveals the intrinsic rate of increase, r , to be proportional to the individual organism's average contribution to the population's productivity, provided crowding effects are negligible. Furthermore, in the very special circumstance where the population's dissipation is not density dependent, i.e., $d = 0$, the carrying capacity, K , is seen to be proportional to an efficiency of some sort, again provided crowding effects are negligible. However, in general, both c and d are non-zero and the carrying capacity does not correspond to an efficiency.

A perhaps better interpretation of K from an ecological energetics point of view can be developed.

Consider an expression for the individual's average contribution to the population's productivity, including crowding effects:

$$\text{individual productivity} = (I_0 - D_0) - (cI_0 - dD_0)N \quad (34)$$

Observe the factor $(cI_0 - dD_0)$, to be called the crowding increment, because each individual's average contribution to the population productivity is diminished by the amount $(cI_0 - dD_0)$ whenever the population size is increased by one member. Indeed, the crowding increment is a measure of the individual's sensitivity to having his average productivity diminished by crowding. Observe next that in equations (33) K is equal to the individual's uncrowded productivity divided by the crowding increment. Thus the carrying capacity equals the individual's average contribution to population productivity under uncrowded conditions divided by a measure of the individual's sensitivity to having that contribution diminished by crowding. Clearly, the smaller the crowding increment, that is, the less sensitive is the individual's average productivity to crowding, then the larger is the carrying capacity.

To conclude, we review some empirical generalizations to which this study lends support. The selection pressure in favor of alleles conferring high- K phenotypes increases, in a seasonal environment, as the dormancy season becomes milder and the growing season longer. Hence proportionately more species and traits responding to K -selection are expected in tropical regions than in temperate regions. In addition, r is proportional to an individual's average productivity under uncrowded conditions, and K is equal to the individual's average uncrowded productivity divided by a measure of sensitivity to crowding. Clearly, a phenotype possessing a high- K might have mediocre productivity in uncrowded conditions offset by a very low sensitivity to having that productivity diminished by crowding. Consequently, one would expect some tropical species to show lower productivity under uncrowded conditions but higher productivity in crowded conditions than comparable temperate species.

In addition, a phenotypically intermediate heterozygote can show a net selective advantage over both homozygotes in a seasonal environment of moderate harshness. If this mechanism for maintaining heterozygosity explains in part the results of electrophoretic analysis of proteins from natural populations, then the loci involved are expected to show fixation for one allele at low latitudes, polymorphism in medium latitudes, and fixation for the other allele at tropical latitudes. However, other causes of heterozygosity might also produce this latitudinal pattern and one would need to confirm that the fitness for the loci involved was density dependent.

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LITERATURE CITED

- Birch, L. C., 1955. Selection in *Drosophila pseudoobscura* in relation to crowding. *Evolution* **9**: 389–399.
- . 1960. The genetic factor in population ecology. *Amer. Natur.* **94**: 5–24.
- Buzzati-Traverso, A. A. 1955. Evolutionary changes in components of fitness and other polygenic traits in *Drosophila melanogaster* populations. *Heredity* **9**: 153–186.
- Gadgil, M., and W. Bossert. 1970. Life historical consequences of natural selection. *Amer. Natur.* **104**: 1–24.
- Haldane, J. B. S. 1932. The causes of evolution. Cornell University Press, Ithaca, N. Y. 235 p.
- Haldane, J. B. S., and S. D. Jayakar. 1963. Polymorphism due to selection of varying direction. *J. Genet.* **58**: 237–242.
- Levins, R. 1968. Evolution in changing environments. Princeton University Press, Princeton, N. J. 120 p.
- Lewontin, R. C., and J. L. Hubby. 1966. A molecular approach to the study of genetic heterozygosity in natural populations. II. Amount of variation and degree of heterozygosity in natural populations of *Drosophila pseudoobscura*. *Genetics* **54**: 595–609.
- Lindeman, R. L. 1942. Trophic dynamic aspect of ecology. *Ecology* **23**: 399–418.
- MacArthur, R. H. 1962. Some generalized theorems of natural selection. *Proc. Nat. Acad. Sci.* **48**: 1893–1897.
- MacArthur, R. H., and E. O. Wilson. 1967. The theory of island biogeography. Princeton University Press, Princeton, N. J. 203 p.
- Smith, J. M. 1968. Mathematical ideas in biology. Cambridge University Press, London. 152 p.
- Wright, S. 1959. Physiological genetics, ecology of populations, and natural selection. *Perspect. Biol. and Med.* **3**: 107–151.